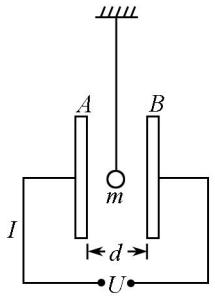
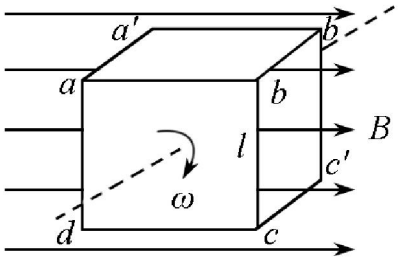
1. the maximum kinetic energy attainable by the moving ball;
2. the mean current in the external circuit after a long time.

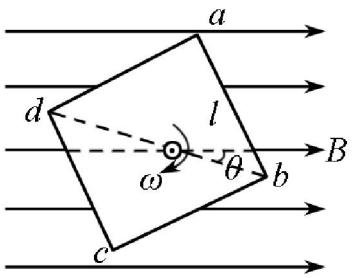


VII. (20 points) In Fig. (a), 12 uniform wire rods form a rigid cube of edge l. Each rod has resistance R. It rotates uniformly about a central axis normal to face $abcd$ at angular speed ω, in a uniform magnetic field of magnitude B normal to that axis. Neglect self-inductance. In Fig. (b), diagonal bd makes angle θ with the field. Find:

1. the currents in rods ba, ad, bc and cd;
2. the external torque needed to maintain uniform rotation.



(a)



(b)